Name_____ Date_____ Period_____

LESSON 7-5 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Plotting Points on the Coordinate Plane

Maryland Standard 6.NS.C.6c · Learning target – I can plot a point with rational coordinates and name the quadrant it lies in.

REMEMBER

- Start at the **origin** (0, 0). Move *x* units right or left, then *y* units up or down.
- Quadrant signs: I (+, +), II (−, +), III (−, −), IV (+, −).
- A point with a 0 coordinate is **on an axis**, not in a quadrant.

WORKED EXAMPLE

Plot (−3.5, 2) and name its quadrant.

Step 1 Start at the origin. Move left 3.5 units.

Step 2 Move up 2 units. Left and up is Quadrant II.

Answer **Quadrant II**

WORD BANK

ordered pair – (x, y): first move right or left, then up or down

origin – (0, 0), where the axes cross

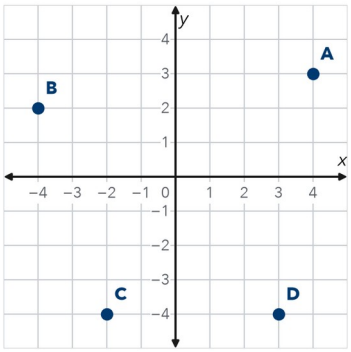
quadrant – one of the four parts of the grid

on an axis – a point with a 0 in it

PRACTICE

show your work · label units

1. Write the ordered pair for each point. Name its quadrant.



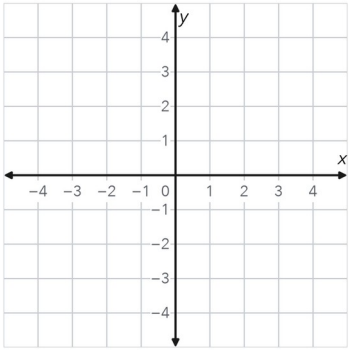
Step 1 Start at the origin. Count left or right to the point: that is x.

Step 2 Count up or down: that is y.

Step 3 Use the signs to name the quadrant.

Point	Ordered pair	Quadrant
A		
B		
C		
D		

2. Town Map. The library is at the origin. Plot and label each place: Park P(2, 4), Pool Q(−4, −1), School S(−3.5, 3), Garden G(4, −2.5).



Step 1 Move right or left first (the first number).

Step 2 Then move up or down (the second number).

Which place is in Quadrant III? _____ Quadrant IV? _____

3. Name the quadrant or axis for each point.

Step 1 Look at the sign of each number.

Step 2 If a number is 0, the point is on an axis.

a. (−7, −2) _____ **b.** (8, −1) _____ **c.** (0, 6) _____

d. (−1.5, 4) _____ **e.** (−9, 0) _____ **f.** (2.5, −6) _____

4. Reasoning. Malia says (−3, 5) and (5, −3) name the same point. Is she right? Explain.

Frame *Malia is ___ because ___. (−3, 5) is in Quadrant ___ and (5, −3) is in Quadrant ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME With someone at home, name a quadrant. Take turns saying an ordered pair that lands in it.

Name_____ Date_____ Period_____

LESSON 7-5 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Reflecting Points Across Axes

Maryland Standard 6.NS.C.6b · Learning target – I can reflect a point across the x-axis or the y-axis and name the new coordinates.

REMEMBER

- **x-axis** reflection: flip the sign of *y*. (3, 4) becomes (3, −4).
- **y-axis** reflection: flip the sign of *x*. (3, 4) becomes (−3, 4).
- Across **both** axes, both signs flip.

WORKED EXAMPLE

Reflect (−2, 5) across the x-axis.

Step 1 Keep x = −2.

Step 2 Change the sign of y: 5 becomes −5.

Answer **(−2, −5)**

WORD BANK

reflection – a mirror flip

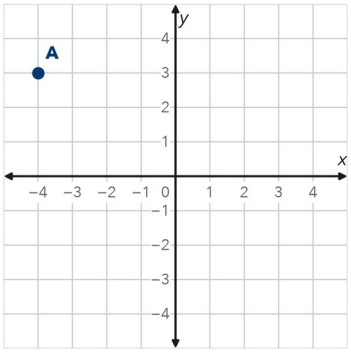
across the x-axis – change the sign of y

across the y-axis – change the sign of x

PRACTICE

show your work · label units

1. Point A is at (−4, 3). Plot point B, the reflection of A across the x-axis. Plot point C, the reflection of A across the y-axis.



Step 1 For B, keep x. Change the sign of y.

Step 2 For C, keep y. Change the sign of x.

a. B = _____

b. C = _____

2. Write the new point after each reflection.

Step 1 Find the mirror: the x-axis or the y-axis.

Step 2 Change the sign of only one number. For both axes, do one flip, then the other.

a. (−6, −2) across the y-axis _____ **b.** (−4, 7) across both axes _____

3. **Park Map.** A bench is the **mirror image** of a bus stop at (7, −2.5). The mirror is the **x-axis**. Where is the bench?

Answer_____

4. **Stretch.** Point K was reflected across the x-axis. The new point is (−6, −5). Where was point K?

K_____

5. Find the Error. Sam reflected (−3, −8) across the y-axis and got (−3, 8). Explain his mistake. Fix it.

Frame *Sam changed the ___-coordinate. Across the y-axis, the ___-coordinate changes sign. The correct point is ___.*

TEST PRACTICE

choose the best answer

6. Point M(−5, 2) is reflected across the y-axis. What is the new point?

A (−5, −2) **B** (5, 2) **C** (5, −2) **D** (2, −5)

7. The points (6, −1) and (6, 1) are reflections of each other across –

A the x-axis **B** the y-axis **C** both axes **D** neither axis

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___